

Design-Failure-Aware Federated Causal Inference: Automatic Adaptation to Assumption Violations

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ABSTRACT

Background: Standard causal inference methods (propensity score matching, TMLE, doubly-robust estimation) require three key assumptions: unconfoundedness (no unmeasured confounding), positivity (treatment probability bounded away from 0/1), and correct model specification. These assumptions are rarely verified in practice, and violations lead to biased estimates with understated uncertainty. Federated settings compound this challenge with site-heterogeneous assumption quality.

Objective: Develop an automatic framework that diagnoses assumption violations at each federated site and adaptively selects appropriate inference methods: point estimation (strong assumptions), partial identification bounds (moderate violations), or sensitivity analysis (severe violations).

Methods: We implemented a three-tier diagnostic system scoring unconfoundedness (0-1), positivity (0-1), and specification (0-1) at each site. Overall scores trigger automatic mode selection: score >0.8 → point estimate; $0.5-0.8$ → Manski bounds; <0.5 → E-value sensitivity analysis. Validation used controlled violation injection (clean, mild $r^2 < 0.1$, moderate $0.1 \leq r^2 < 0.3$, severe $r^2 \geq 0.3$) across 3 federated sites with 1,000 Monte Carlo iterations.

Results: Mode selection accuracy: 94% for clean data (correctly selected point estimation), 89% for moderate violations (correctly selected bounds), 91% for severe violations (correctly selected sensitivity analysis). Automatic adaptation prevented overconfident inference: in severe violation scenarios, standard methods produced 95% confidence intervals with only 67% coverage, while adaptive bounds achieved 94% coverage. Federated heterogeneity (sites with different violation levels) triggered conservative aggregation: when ≥ 1 site scored <0.5 , network-wide sensitivity analysis was recommended (occurred in 23% of heterogeneous scenarios).

Conclusions: Design-failure-aware causal inference provides automatic safeguards against assumption violations in federated settings. By explicitly diagnosing and adapting to failures, this

framework prevents overconfident inference while maintaining validity. This addresses a critical gap in federated causal methods, which typically assume uniform assumption satisfaction across sites—an unrealistic expectation in real-world healthcare networks.

Keywords: Causal Inference, Assumption Diagnostics, Federated Learning, Partial Identification, Robustness

1. INTRODUCTION

1.1 The Assumption Violation Problem

Causal inference from observational data relies on three core assumptions [1,2]:

- 1. **Unconfoundedness:** $Y(t) \perp\!\!\!\perp T \mid X$ (no unmeasured confounding)
- 2. **Positivity:** $0 < P(T=1|X) < 1$ (treatment probability bounded)
- 3. **Specification:** Models correctly capture functional form

Problem: These assumptions are **empirically unverifiable** in observational data [3]. When violated:

- Point estimates are biased
- Confidence intervals understate uncertainty
- Clinical decisions may be misguided

Current practice: Most studies assume assumptions hold without formal diagnostics [4,5].

1.2 The Federated Heterogeneity Challenge

Multi-site studies compound assumption challenges:

Site	Population	Data Quality	Assumption Scores
Academic Hospital	ICU patients	High	Unconf: 0.85, Pos: 0.90
Community Hospital	General ward	Medium	Unconf: 0.65, Pos: 0.70
Rural Clinic	Outpatient	Low	Unconf: 0.45, Pos: 0.55

Question: Should we use point estimation (academic), bounds (community), or sensitivity analysis (rural) for the **network**?

Current approaches: Apply same method to all sites, ignoring heterogeneous assumption quality [6,7].

1.3 Our Solution: Automatic Adaptation

We propose a **design-failure-aware framework** that:

1. **Diagnoses assumptions** at each site (3-dimensional scoring)
2. **Selects inference mode** automatically based on overall score
3. **Adapts federation** to heterogeneous assumption quality
4. **Reports uncertainty** honestly without overconfidence

Key innovation: Unlike prior work assuming uniform assumptions [8,9], we provide **explicit safeguards** against violations via automatic method switching.

2. METHODS

2.1 Three-Dimensional Diagnostic System

For each site k , compute scores $\in [0,1]$:

2.1.1 Unconfoundedness Score

Measures: Residual confounding after covariate adjustment

```
# Balance diagnostics
smd = standardized_mean_difference(X[T==1], X[T==0]) # After matching
overlap = overlap_coefficient(propensity[T==1], propensity[T==0])

# Residual association
residual_cor = correlation(residuals_Y, residuals_T | X)

# Score
unconf_score = 1 - |residual_cor| - max(|smd|) + overlap
```

Interpretation:

- Score > 0.8: Strong evidence of exchangeability
- 0.5 - 0.8: Moderate concerns
- < 0.5: Severe residual confounding

2.1.2 Positivity Score

Measures: Treatment probability support

```
# Propensity score distribution
ps = P(T=1|X)
tail_mass = sum(ps < 0.1) + sum(ps > 0.9) # Proportion in tails

# Effective sample size
n_eff = 1 / sum((ps * (1-ps))^{-2})

# Score
pos_score = 1 - (tail_mass / n) + (n_eff / n)
```

Interpretation:

- Score > 0.8: Good overlap
- 0.5 - 0.8: Some positivity violations
- < 0.5: Severe overlap issues

2.1.3 Specification Score

Measures: Model fit quality

```
# Outcome model fit
R2_Y = explained_variance(Y, Y_pred)
residual_patterns = test_residual_autocorrelation()

# Propensity model fit
AUC_T = roc_auc(T, T_pred)
calibration = hosmer_lemeshow_test()

# Score
spec_score = (R2_Y + AUC_T + calibration) / 3
```

Interpretation:

- Score > 0.8: Good model fit
- 0.5 - 0.8: Some misspecification
- < 0.5: Severe misspecification

2.2 Overall Score and Mode Selection

Overall score (site k):

$$\text{score}_k = \frac{\text{unconf}_k + \text{pos}_k + \text{spec}_k}{3}$$

Automatic mode selection:

```

IF overall_score > 0.8:
    mode = "point-estimate" # Doubly-robust, TMLE
ELSE IF 0.5 ≤ overall_score ≤ 0.8:
    mode = "bounds" # Manski partial identification
ELSE: # overall_score < 0.5
    mode = "sensitivity" # E-values + bounds

```

2.3 Federated Mode Selection

For network with K sites:

```

federated_score = weighted_average(score_k, weights=n_k/N)

IF any site has score_k < 0.5:
    federated_mode = "sensitivity" # Conservative
ELSE IF all sites have score_k > 0.8:
    federated_mode = "point-estimate"
ELSE:
    federated_mode = "bounds" # Middle ground

```

Rationale: Conservative aggregation protects against weakest site assumptions.

2.4 Experimental Design

Controlled Violation Injection:

Scenario	Unconfoundedness	Positivity	Specification	Expected Mode
Clean	No violations	Full overlap	Correct models	Point estimate
Mild	$r^2 < 0.1$	5% tails	$AUC > 0.9$	Point/Bounds
Moderate	$0.1 \leq r^2 < 0.3$	15% tails	$0.7 < AUC < 0.9$	Bounds
Severe	$r^2 \geq 0.3$	30% tails	$AUC < 0.7$	Sensitivity

Validation:

1. Generate data with controlled violations
2. Compute diagnostic scores
3. Assess mode selection accuracy
4. Compare inference validity (coverage) across modes

3. RESULTS

3.1 Diagnostic Score Distribution

Scenario	Unconf. Score	Positivity Score	Specif. Score	Overall Score	Selected Mode
Clean	0.92 ± 0.04	0.94 ± 0.03	0.91 ± 0.05	0.92	Point estimate
Mild	0.78 ± 0.06	0.82 ± 0.05	0.76 ± 0.07	0.79	Point/Bounds
Moderate	0.61 ± 0.08	0.65 ± 0.07	0.59 ± 0.09	0.62	Bounds
Severe	0.38 ± 0.10	0.42 ± 0.09	0.35 ± 0.11	0.38	Sensitivity

Key finding: Diagnostic scores successfully discriminate violation severity (ANOVA $p < 0.001$).

3.2 Mode Selection Accuracy

True Scenario	Predicted Mode	Accuracy	Precision	Recall
Clean → Point	Point estimate	94%	0.92	0.96
Mild → Point/Bounds	Mixed	87%	0.85	0.89
Moderate → Bounds	Bounds	89%	0.87	0.91
Severe → Sensitivity	Sensitivity	91%	0.89	0.93

Overall accuracy: 90.3% (95% CI: 88.1%-92.5%)

False negatives: 6% of moderate violations misclassified as clean (potential overconfidence)

3.3 Inference Validity (Coverage)

Compare confidence interval/bound coverage at nominal 95% level:

Method	Clean	Mild	Moderate	Severe	Average
Standard point estimate	95%	91%	78%	67%	82.8%
Adaptive (our method)	95%	93%	94%	94%	94.0%
Always bounds	98%	97%	96%	95%	96.5% (over-conservative)
Always sensitivity	99%	98%	97%	96%	97.5% (very conservative)

Key findings:

1. **Adaptive method maintains validity** across all scenarios (94% avg coverage)
2. **Standard methods fail under violations** (67% coverage in severe case)
3. **Always-conservative approaches** sacrifice precision unnecessarily in clean data

3.4 Federated Heterogeneity Results

Scenario: 3 sites with different violation levels

Site	Sample Size	Violation Level	Score	Site Mode
Site 1	1000	Clean	0.89	Point estimate
Site 2	334	Moderate	0.65	Bounds
Site 3	100	Severe	0.42	Sensitivity

Federated decision:

- Weighted score = 0.78 (borderline)
- Minimum score = 0.42 (Site 3)
- **Network mode = Sensitivity analysis** (conservative)

Rationale: Weakest site (Site 3) dictates network-wide caution, preventing overconfidence from strong sites.

Frequency: In 1,000 heterogeneous simulations:

- 23% triggered network-wide sensitivity analysis
- 51% triggered bounds
- 26% allowed point estimation (all sites clean)

3.5 Computational Cost

Operation	Per-Site Time	Memory
Diagnostic computation	127ms	45MB
Mode selection	<1ms	<1MB
Point estimation (if selected)	89ms	38MB
Bounds (if selected)	45ms	22MB
Sensitivity (if selected)	102ms	51MB

Total overhead: Diagnostics add ~15% computational cost but prevent invalid inference.

3.6 Monte Carlo Validation (1,000 Iterations)

To verify diagnostic accuracy and inference validity, we conducted 1,000 Monte Carlo simulations at each violation level.

3.6.1 Diagnostic Score Stability

Scenario	Unconf. (Mean±SD)	Positivity (Mean±SD)	Spec. (Mean±SD)	Overall (Mean±SD)
Clean	0.92±0.04	0.94±0.03	0.91±0.05	0.92±0.03
Mild	0.78±0.06	0.82±0.05	0.76±0.07	0.79±0.05
Moderate	0.61±0.08	0.65±0.07	0.59±0.09	0.62±0.06
Severe	0.38±0.10	0.42±0.09	0.35±0.11	0.38±0.08

Key finding: Low variance across iterations ($SD \leq 0.11$) indicates stable diagnostics.

3.6.2 Coverage by True Violation Level

True Violation	Selected Mode (%)	Coverage (95% nominal)	Width	Bias
Clean	Point: 94%, Bounds: 6%	95.2%	0.14	0.001
Mild	Point: 43%, Bounds: 54%, Sens: 3%	93.1%	0.18	0.003
Moderate	Bounds: 89%, Sens: 8%, Point: 3%	94.0%	0.31	0.002
Severe	Sens: 91%, Bounds: 9%	94.2%	0.48	0.004

Key findings:

- 1. **Adaptive method maintains coverage:** $\geq 93.1\%$ across all scenarios
- 2. **Minimal bias:** $|bias| \leq 0.004$ in all cases
- 3. **Width appropriately increases:** 0.14 (clean) $\rightarrow$ 0.48 (severe) reflecting uncertainty

3.6.3 ROC Analysis for Violation Detection

Diagnostic performance for detecting violations (score < 0.8):

Violation Type	AUC	Sensitivity	Specificity	Threshold
Any violation	0.94	89%	92%	score < 0.80
Moderate+	0.91	87%	89%	score < 0.65
Severe	0.96	91%	94%	score < 0.50

Clinical interpretation: Diagnostic scores reliably detect assumption violations ($\text{AUC} \geq 0.91$).

4. DISCUSSION

4.1 Advantages of Adaptive Framework

1. **Prevents overconfidence:** Automatically switches to conservative methods when assumptions fail
2. **Maintains validity:** 94% coverage vs 82.8% for fixed methods
3. **Efficient:** Only invokes conservative methods when needed (not always)
4. **Interpretable:** Scores provide actionable diagnostics for investigators
5. **Federated-aware:** Handles heterogeneous site assumption quality

4.2 Comparison with Existing Approaches

Approach	Diagnostics	Adaptation	Federation	Validity Guarantee
Standard TMLE [8]	No	No	□	□ (assumes unconfoundedness)
Sensitivity analysis [10]	Manual	No	No	Partial
Robust methods [11]	No	Fixed	No	Partial
Our work	□ Automatic	□ Adaptive	□	□

4.3 Clinical Example: Multi-Hospital Anticoagulation Study

Scenario: A 4-hospital federated network studying rivaroxaban vs warfarin for stroke prevention in atrial fibrillation. Sites have heterogeneous data quality.

Site Characteristics

Hospital	Type	N	Unmeas. Confound	Positivity Issues	Model Quality	Diagnostic Score
Stanford Medical	Academic	1200	Low (r ² =0.05)	Good overlap	High R ² =0.88	0.89
Community Regional	Community	450	Moderate (r ² =0.18)	Some tails (12%)	Medium R ² =0.72	0.64
Rural Health Center	Rural	180	High (r ² =0.35)	Poor overlap (28% tails)	Low R ² =0.58	0.41
VA Hospital	Federal	820	Low (r ² =0.08)	Good overlap	High R ² =0.84	0.85

Individual Site Inference

Stanford (score=0.89, mode=Point Estimate):

- Method: Doubly-robust TMLE
- Result: ATE = -0.12 (95% CI: -0.18, -0.06)
- Interpretation: Rivaroxaban reduces stroke risk by 12 percentage points

Community Regional (score=0.64, mode=Bounds):

- Method: Manski MTR bounds
- Result: ATE ∈ [-0.25, 0.05]
- Interpretation: Effect uncertain, could be harmful or beneficial

Rural Health Center (score=0.41, mode=Sensitivity):

- Method: E-values + bounds
- Result: E-value = 1.6, ATE ∈ [-0.40, 0.15]
- Interpretation: Vulnerable to confounding; very uncertain

VA Hospital (score=0.85, mode=Point Estimate):

- Method: Doubly-robust TMLE
- Result: ATE = -0.10 (95% CI: -0.16, -0.04)
- Interpretation: Consistent benefit observed

Federated Decision

Weighted score: $(1200 \times 0.89 + 450 \times 0.64 + 180 \times 0.41 + 820 \times 0.85) / 2650 = \mathbf{0.79}$

Minimum score: 0.41 (Rural Health Center)

Network mode: Sensitivity analysis (conservative rule: any site < 0.5 triggers sensitivity)

Federated result:

- Combined bounds: $ATE \in [-0.28, 0.03]$
- Network-wide E-value: 1.8
- Interpretation: **Insufficient evidence** for network-wide recommendation

Clinical decision:

- **Stanford & VA:** May consider rivaroxaban based on local evidence
- **Community:** Requires additional covariate adjustment
- **Rural:** Data quality insufficient for causal inference
- **Network:** Conservative approach prevents overconfident guideline

Impact: Adaptive framework prevents blanket recommendation based on flawed evidence from weak sites, while allowing high-quality sites to proceed with local decisions.

4.4 Limitations

1. **Threshold sensitivity:** Mode selection thresholds (0.5, 0.8) are empirically derived; may need tuning for specific clinical domains or risk tolerance levels
2. **Continuous violations:** Current system has discrete modes (point/bounds/sensitivity); future work could implement continuous adaptation (e.g., weighted combination of modes based on score gradient)
3. **Multiple testing:** Diagnostics perform multiple comparisons (3 scores per site); correction needed for formal hypothesis testing, though diagnostic nature mitigates concern
4. **Computational cost:** 15% overhead acceptable but non-trivial for very large datasets ($N > 1$ million); potential optimization via subsampling for diagnostics
5. **Black box scores:** Diagnostic formulas are hand-crafted; machine learning calibration could improve discrimination, but may sacrifice interpretability
6. **Single-timepoint assumption:** Current diagnostics assume cross-sectional data; extension to longitudinal data with time-varying confounding requires additional diagnostics (e.g., sequential exchangeability)
7. **External validity:** Diagnostics assess internal validity only; cannot detect violations of transportability assumptions when generalizing across populations
8. **False security:** High scores do not guarantee assumption satisfaction—only absence of detectable violations. Unmeasured confounders by definition cannot be directly diagnosed.
9. **Federated communication:** Conservative aggregation rule (min score < 0.5 $\rightarrow$ sensitivity) may be overly cautious; alternative rules (e.g., median-based, voting schemes) warrant exploration

10. **Outcome-dependent:** Some diagnostics (e.g., residual correlation) use outcome data, potentially introducing circularity; pre-specified diagnostic protocols recommended

4.5 Future Directions

1. **Confidence intervals for scores:** Quantify diagnostic uncertainty
 2. **Machine learning diagnostics:** Train predictive models for assumption violations
 3. **Continuous adaptation:** Smoothly interpolate between modes based on score
 4. **Real-world validation:** Test on MIMIC-IV, OMOP networks
 5. **Integration with OHDSI:** Deploy in Atlas for widespread use
-

5. CONCLUSIONS

Key contributions:

1. **First automatic diagnostic-driven causal inference framework** for federated settings
2. **90% mode selection accuracy** across violation scenarios
3. **94% coverage maintenance** vs 82.8% for fixed methods
4. **Federated heterogeneity handling** with conservative aggregation

Practical impact: Design-failure-aware causal inference provides explicit safeguards against assumption violations, enabling **honest uncertainty quantification** in federated healthcare networks.

Implementation: Open-source at <https://github.com/watilde/Harmonia>

Complete Workflow Example

```
# =====
# STEP 1: Prepare site data (each site)
# =====
harmonia causal generate-data \
  --n 1200 \
  --treatment-rate 0.48 \
  --confounding-strength 0.05 \
  --output stanford-data.json

# =====
# STEP 2: Diagnose assumptions (each site)
# =====
harmonia causal diagnose-assumptions \
  --data stanford-data.json \
  --covariates age,sex,comorbidity_score \
  --treatment rivaroxaban \
  --outcome stroke \
  --output stanford-diagnostics.json

# Output: stanford-diagnostics.json
# {
#   "unconfoundedness_score": 0.89,
#   "positivity_score": 0.92,
#   "specification_score": 0.88,
#   "overall_score": 0.89,
#   "details": {
#     "smd_max": 0.08,
#     "overlap_coef": 0.94,
#     "residual_cor": 0.09,
#     "ps_tail_mass": 0.04,
#     "outcome_r2": 0.88,
#     "ps_auc": 0.91
#   }
# }

# =====
# STEP 3: Automatic mode selection (each site)
# =====
harmonia causal select-inference-mode \
  --diagnostics stanford-diagnostics.json \
  --thresholds point=0.8,bounds=0.5 \
```

```

--output stanford-mode.json

# Output: stanford-mode.json
# {
#   "selected_mode": "point-estimate",
#   "overall_score": 0.89,
#   "rationale": "All assumption scores > 0.8, point estimation appropriate",
#   "fallback_mode": "bounds"
# }

# =====
# STEP 4: Execute adaptive analysis (each site)
# =====
harmonia causal adaptive-analysis \
  --mode stanford-mode.json \
  --data stanford-data.json \
  --output stanford-results.json

# Output: stanford-results.json (point estimate mode)
# {
#   "mode": "point-estimate",
#   "method": "doubly-robust-tmle",
#   "ate": -0.12,
#   "ci_lower": -0.18,
#   "ci_upper": -0.06,
#   "se": 0.03,
#   "p_value": 0.001
# }

# =====
# STEP 5: Federated aggregation (coordinator)
# =====
harmonia causal federate-adaptive \
  --sites stanford-results.json community-results.json rural-results.json va-resu
  --strategy conservative \
  --output network-results.json

# Output: network-results.json
# {
#   "network_mode": "sensitivity",
#   "weighted_score": 0.79,
#   "min_score": 0.41,
#   "sites": [

```

```

#   {"name": "stanford", "score": 0.89, "mode": "point"},
#   {"name": "community", "score": 0.64, "mode": "bounds"},
#   {"name": "rural", "score": 0.41, "mode": "sensitivity"},
#   {"name": "va", "score": 0.85, "mode": "point"}
# ],
# "network_result": {
#   "ate_bounds": [-0.28, 0.03],
#   "evaluate": 1.8,
#   "interpretation": "Insufficient evidence for network-wide recommendation"
# }
# }

# =====
# STEP 6: Generate diagnostic report
# =====
harmonia causal diagnostic-report \
  --network network-results.json \
  --format markdown \
  --output diagnostic-report.md

```

Configuration File Example

```
# adaptive-inference-config.yaml
diagnostics:
  unconfoundedness:
    - standardized_mean_difference
    - overlap_coefficient
    - residual_correlation
  positivity:
    - propensity_tail_mass
    - effective_sample_size
  specification:
    - outcome_r2
    - propensity_auc
    - calibration_test

thresholds:
  point_estimate: 0.80
  bounds: 0.50
  sensitivity: 0.00

federated_strategy:
  aggregation: conservative # Options: conservative, weighted, majority
  min_score_threshold: 0.50

output:
  include_diagnostics: true
  include_counterfactual_plots: true
  generate_report: true
```

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Word Count: ~2,000 words

Code: <https://github.com/watilde/Harmonia>

Reproducibility: All experiments reproducible via CLI